

Q7	
(a)(i)	When a graph showing the variation of count-rate from a radioactive sample over time is plotted, we observe fluctuations instead of a smooth curve.
(a)(ii)	Variations in external factors such as temperature and pressure do not affect the curve variation in the count-rate over time
(b)(i)	From the figure, at $t = 0$, $C_0 = 475 \text{ s}^{-1}$. When the count rate is halved, $C_t = \frac{475}{2} = 237.5 \text{ s}^{-1}$. To reach this value, reading from the graph, half-life is 15 days At $t = 10$ days, $C = 300 \text{ s}^{-1}$. This is halved at $C = 150 \text{ s}^{-1}$, which is at $t = 25$ days. This also gives a half-life of 15 days. The (averaged) half life is thus 15 days.
(b)(ii)	Beta particles have a sufficiently long range that could cross the entire lab, so the student is exposed to it even when not conducting the experiment. OR Beta particles are ionising radiation that can cause health problems like cancer when exposed to a sufficiently high dose. Radiation dosage looks at the total amount accumulated over time and if this is high enough due to prolonged exposure, it can be a health hazard.
(b)(iii)	If the product is not stable, it can decay and that will contribute to the count rate measured which will not be just due to the decay of phosphorous-32 alone.

